Status	Finished
Started	Monday, 5 October 2026, 10:22 AM
Completed	Monday, 5 October 2026, 10:28 AM
Duration	5 mins 41 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Correct

Mark 1.00 out of 1.00

What is the first step in binary search?

a. Divide the list into two equal parts
b. Compare the target element with the first element in the list
c. Sort the list
d. Compare the target element with the middle element in the list ✓

Question **2**

Correct

Mark 1.00 out of 1.00

In binary search, how is the middle element determined?

a. By dividing the list length by two ✓
b. By comparing each element sequentially
c. By starting from the first element
d. By using a hash function

Question **3**

Correct

Mark 1.00 out of 1.00

Which of the following best describes the process of a linear search?

a. Sorting the list before searching
b. Skipping every second element
c. Checking each element sequentially ✓
d. Dividing the list in half repeatedly

Question **4**

Correct

Mark 1.00 out of 1.00

In which type of search is the list divided into smaller sublists during the search process?

a. Linear search
b. Sequential search
c. Hash search
d. Binary search ✓

Question **5**

Incorrect

Mark 0.00 out of 1.00

What happens in a binary search if the list has an even number of elements?

a. The higher middle element is chosen as the middle element
b. The middle element is chosen randomly
c. The lower middle element is chosen as the middle element ✗
d. The search stops

Question **6**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

a. 10 ✓
b. 500
c. 1000
d. 100

Question **7**

Correct

Mark 1.00 out of 1.00

In binary search, if the target element is less than the middle element, where does the search continue?

a. In the entire list
b. In the right sublist
c. At the beginning of the list
d. In the left sublist ✓

Question **8**

Correct

Mark 1.00 out of 1.00

In linear search, how is the element searched?

- a. By comparing each element in the list sequentially ✓
- b. By sorting the list first
- c. By dividing the list into halves
- d. By using a hash function

Question **9**

Correct

Mark 1.00 out of 1.00

What is the key characteristic of binary search?

- a. It works on unsorted lists
- b. It can be applied only if the list is sorted ✓
- c. It compares elements sequentially
- d. It always starts from the beginning of the list

Question **10**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

- a. log n
- b. n/2
- c. n ✓
- d. 1

Question **11**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

- a. O👎
- b. O(n log n)
- c. O(log n) ✓
- d. O(1)

Question **12**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

a. The first element is returned
b. The last element is returned
c. The search is considered unsuccessful ✓
d. An error is raised

Question **13**

Correct

Mark 1.00 out of 1.00

Finding the location of a given item in a collection of items is called

a. Discovering
b. Finding
c. Searching ✓
d. Mining

Question **14**

Correct

Mark 1.00 out of 1.00

Which of the following is a type of searching method?

a. Merge search
b. Linear search ✓
c. Bubble search
d. Quick search

Question **15**

Correct

Mark 1.00 out of 1.00

What happens when the element is found in linear search?

a. The search starts over from the beginning
b. The search stops immediately ✓
c. The search continues until the end of the list
d. The search backtracks to find duplicate elements